

Total marks (120)
Attempt Questions 1 – 10
All questions are of equal value

Answer each question in a SEPARATE writing booklet.

QUESTION 1	(12 MARKS)	Use a SEPARATE writing booklet	Marks
(a)	$f(x) = 2x^3 - 3x$. Evaluate $f(-3)$.		1
(b)	Simplify $2 \times -5 - -12 $.		1
(c)	Simplify $\frac{4x^2 - 9}{4x + 6}$.		2
(d)	Rationalise the denominator and simplify $\frac{\sqrt{3}-1}{\sqrt{3}+1}$.		2
(e)	Find the values of $\frac{\tan x}{x+1}$ when $x = 2.3$ radians, giving your answer correct to two decimal places.		2
(f)	The length of a rectangle is increased by 5% and the width is decreased by 2%. Find the percentage increase in its area.		2
(g)	Solve the inequality $2x^2 + 7x - 4 \geq 0$.		2

QUESTION 2 (12 MARKS) Use a SEPARATE writing booklet

Marks

(a) Differentiate the following functions:

(i) $3x^2 - 2\sqrt{x}$. **2**

(ii) $\frac{1}{(3x-2)^2}$. **2**

(b) Find $\int (e^{2x} - \sin x) dx$. **2**

(c) Evaluate $\int_4^5 \frac{1}{x-3} dx$ correct to two decimal places. **2**

(d) (i) Differentiate $x^2 \ln x$. **2**

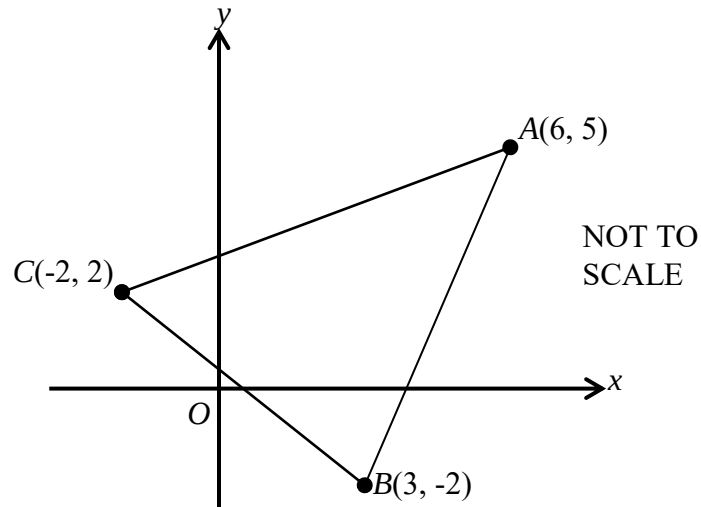
(ii) Hence find $\int x \ln x dx$. **2**

QUESTION 3

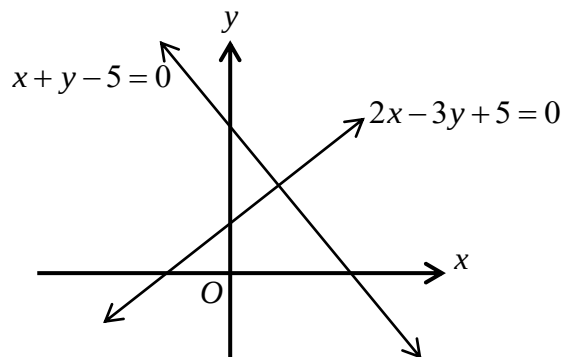
(12 MARKS) Use a SEPARATE writing booklet

Marks

- (a) The diagram shows the three points $A(6, 5)$, $B(3, -2)$ and $C(-2, 2)$.



- | | | |
|-------|--|---|
| (i) | Find the length of AC . | 1 |
| (ii) | Show that the equation of AC is $3x - 8y + 22 = 0$. | 2 |
| (iii) | Find the perpendicular distance from B to AC . | 2 |
| (iv) | Hence find the area of $\triangle ABC$. | 2 |
- (b) The diagram shows the graphs of the lines $x + y - 5 = 0$ and $2x - 3y + 5 = 0$.



- | | | |
|------|--|---|
| (i) | Find the coordinates of the point of intersection of the two lines. | 3 |
| (ii) | Copy the diagram into your writing booklet, and shade the region which satisfies both the inequalities $x + y - 5 \leq 0$ and $2x - 3y + 5 \leq 0$. | 2 |

QUESTION 4 (12 MARKS) Use a SEPARATE writing booklet **Marks**

(a) Solve the equation $2^{2x} - 7 \times 2^x - 8 = 0$. **3**

(b) (i) Write down the discriminant of the expression $kx^2 + 4x + k$. **1**

(ii) For what values of k does the equation $kx^2 + 4x + k = 0$ have no real roots? **2**

(c) The equation of a parabola is $x^2 = -8(y + 2)$.

(i) Write down the coordinates of the vertex of the parabola. **1**

(ii) What is the focal length of the parabola? **1**

(iii) Write down the coordinates of the focus of the parabola. **1**

(iv) What is the equation of the directrix of the parabola? **1**

(d) If α, β are the roots of the quadratic equation $ax^2 + bx + c = 0$, prove that **2**

$$(\alpha - \beta)^2 = \frac{b^2 - 4ac}{a^2}.$$

QUESTION 5

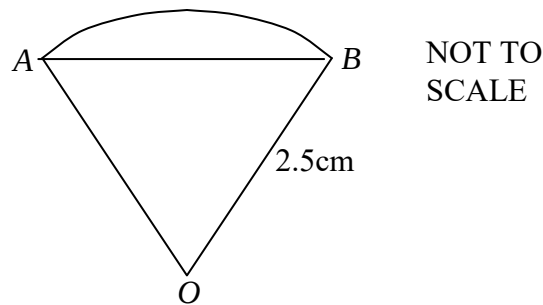
(12 MARKS)

Use a SEPARATE writing booklet

Marks

- (a) In $\triangle PQR$, $PQ = 12\text{cm}$, $QR = 8\text{cm}$, and the area of the triangle is 28.8 cm^2 . 3
Find two possible values for the size of $\angle PQR$ (to the nearest degree).

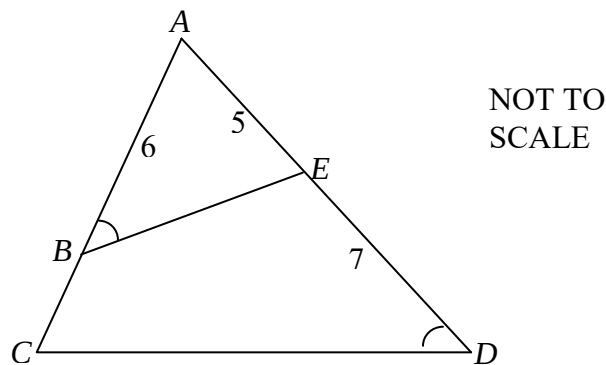
(b)



The curve AB is the arc of a circle, centre O and radius 2.5cm .
The length of the arc is 4cm .

- (i) Show that $\angle AOB = 1.6$ radians. 1
- (ii) Find the length of the chord AB (to one decimal place). 2
- (iii) Find the area of the minor segment formed by arc AB and the chord AB . 2

(c)



In the diagram, $\angle ABE = \angle ADC$, $AE = 5\text{cm}$, $DE = 7\text{cm}$, $AB = 6\text{cm}$.

- (i) Prove that $\triangle ABE$ is similar to $\triangle ACD$. 2
- (ii) Hence find the length of BC . 2

QUESTION 6 (12 MARKS) Use a SEPARATE writing booklet **Marks**

- (a) In an arithmetic series, the sum of the first 16 terms is 288 and the sixth term is 8. **4**
Find the first three terms of the series.
- (b) Find the least positive integer n that satisfies the inequation **3**
$$4 \times (1.2)^n > 560.$$
- (c) \$1500 is deposited in an account at the start of each year, the first deposit being in 2006 and the last in 2015.
The account pays interest at 8% p.a., compounded half-yearly.
- (i) Find the value of the first deposit at the end of 2015. **2**
- (ii) Find the total value of the account at the end of 2015. **3**

QUESTION 7**(12 MARKS)**

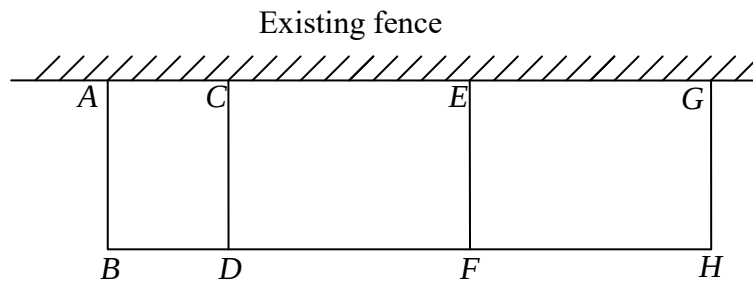
Use a SEPARATE writing booklet

Marks

- (a) Find the equation of the normal to the curve $y = 3x^2 - 8x + 2$ at the point $(2, -2)$ on the curve. **3**

- (b) Show that the curve $y = x^3 - 12x^2 + 48x + 50$ has only one stationary point, and show that it is a horizontal point of inflexion. **4**

(c)



A rectangular yard is to be constructed using an existing fence as one side, and the yard is to be divided into three rectangular regions, as shown in the diagram.

40 metres of fencing is to be used to construct the five lengths of fencing, that is: AB , CD , EF , GH , and BH .

- (i) If the length of AB is x metres, find the length of BH as a function of x . **1**
- (ii) Hence find the dimensions of the yard so that the total area is a maximum. **4**

QUESTION 8 (12 MARKS) Use a SEPARATE writing booklet **Marks**

- (a) (i) State the period of the function $y = 1 + \sin 2x$. **1**
- (ii) Hence sketch the graph of $y = 1 + \sin 2x$ for $-\pi \leq x \leq 2\pi$. **3**
- (b) If $\cos \theta = -\frac{2}{3}$ and $\tan \theta > 0$, find the exact value of $\sin \theta$ (without finding the value of θ). **2**
- (c) (i) Sketch the graph of $y = \cos x$ for $0 \leq x \leq 2\pi$. **1**
- (ii) Hence solve the inequation $\cos x \leq \frac{\sqrt{3}}{2}$ for $0 \leq x \leq 2\pi$. **2**
- (d) (i) Given the trigonometric identity $\sin^2 \theta + \cos^2 \theta = 1$, what operation will produce the identity $\tan^2 \theta + 1 = \sec^2 \theta$? **1**
- (ii) Hence find $\int (\tan^2 x) dx$. **2**

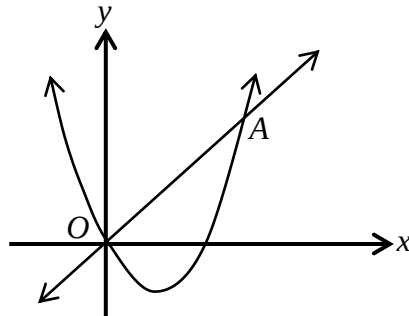
QUESTION 9

(12 MARKS) Use a SEPARATE writing booklet

Mark

- (a) The arc of the graph of $y = 4 - x^2$ between $(0, 4)$ and $(3, -5)$, is rotated about the y axis. Find the volume of the solid formed. **3**

(b)



The line $y = 2x$ and the parabola $y = x(x - 3)$ meet at $O(0, 0)$ and A .

- (i) Find the x coordinate of A . **1**
- (ii) Hence find the area bounded by the line and the parabola. **3**
- (c) A function $y = f(x)$ has the following function values:

x	0	2	4	6	8
$f(x)$	3.2	1.4	0.6	2.2	3.6

- (i) Use Simpson's Rule with five function values to find the approximate value of

$$\int_0^8 f(x) dx.$$

- (ii) The graph of $y = f(x)$ for $0 \leq x \leq 8$ is rotated about the x -axis. Use Simpson's Rule and five function values to find the approximate volume of the solid formed. **3**

QUESTION 10 (12 MARKS) Use a SEPARATE writing booklet

Marks

- (a) The number of bacteria N in a colony after t minutes, is given by

$$N = 2000e^{0.005t}$$

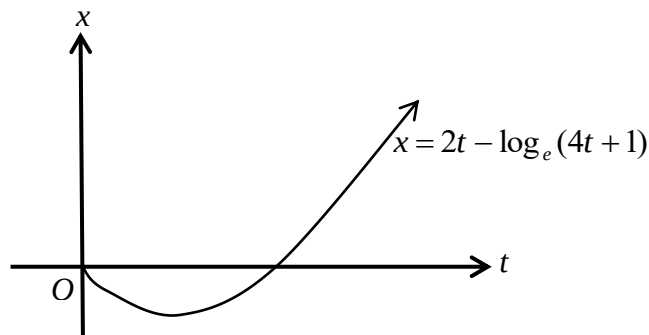
- (i) Find the number of bacteria when $t = 10$. **1**
- (ii) Find the rate at which the colony is increasing when $t = 10$. **2**
- (iii) Find the time taken for the population to double. **1**

- (b) A particle moves along a straight line. Its position x metres from O , t seconds after starting, is given by $x = 2t - \log_e(4t + 1)$

- (i) Show that the velocity, v m/sec, and the acceleration, a m/sec², are : **2**

$$v = \frac{8t - 2}{4t + 1} \quad \text{and} \quad a = \frac{16}{(4t + 1)^2}.$$

- (ii) Find the initial velocity and the initial acceleration. **2**
- (iii) Find when the velocity is zero. **1**
- (iv)



The diagram shows the displacement-time graph.

On separate number planes, showing relevant features, draw:

- (α) the velocity – time graph **2**
- (β) the acceleration – time graph. **1**

End of paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

TRIAL H.S.C.

EXAMINATION
2006

"MATHEMATICS"

SOLUTIONS AND MARKERS
COMMENTS.

Mathematics: Question 1

Suggested Solutions

Marks
Awarded

Marker's Comments

$$\begin{aligned} (a) \quad f(x) &= 2x^3 - 3x \\ f(-3) &= 2(-3)^3 - 3(-3) \\ &= -45 \end{aligned}$$

(1)

1

No problems

$$\begin{aligned} (b) \quad 2 \times |-5| - |-12| &= 10 - 12 \\ &= -2 \end{aligned}$$

(1)

1

No problems

$$\begin{aligned} (c) \quad \frac{4x^2 - 9}{4x + 6} &= \frac{(2x-3)(2x+3)}{2(2x+3)} \\ &= \frac{2x-3}{2} \end{aligned}$$

(2)

1

Correct factoring

1

Correct simplify

$$\begin{aligned} (d) \quad \frac{\sqrt{3}-1}{\sqrt{3}+1} &= \frac{\sqrt{3}-1}{\sqrt{3}+1} \times \frac{\sqrt{3}-1}{\sqrt{3}-1} \\ &= \frac{3 - 2\sqrt{3} + 1}{3 - 1} \\ &= \frac{4 - 2\sqrt{3}}{2} \\ &= 2 - \sqrt{3} \end{aligned}$$

(2)

1

Correct first step.

Many did not
Simplify Fully.

$$\begin{aligned} (e) \quad \frac{\tan x}{x+1} &= \frac{\tan 2.3}{2.3+1} \quad \text{NB Radians} \\ &= -0.34 \text{ (2 d.p.)} \end{aligned}$$

(2)

1

Correct substitution

1

Correct rounding.

$$\begin{aligned} (f) \quad \text{New area} &= (1.05x)(0.98y) \\ &= 1.029xy \end{aligned}$$

where x, y are length and breadth
of original rectangle.

$\therefore$ Area increases by 2.9%

(2)

1

Correct interpretation

Many did not
indicate correct
increase of 2.9%.

$$\begin{aligned} (g) \quad 2x^2 + 7x - 4 &\geq 0 \\ (2x-1)(x+4) &\geq 0 \\ x &\leq -4, \quad x \geq \frac{1}{2} \end{aligned}$$

(2)

1

Correct factoring

1

Correct values
AND inequality
signs.



Mathematics: Question 2

Suggested Solutions	Marks Awarded	Marker's Comments
(a) (i) $\frac{d}{dx} (3x^2 - 2\sqrt{x}) = \frac{d}{dx} (3x^2 - 2x^{\frac{1}{2}})$ $= 6x - x^{-\frac{1}{2}}$ OR $6x - \frac{1}{\sqrt{x}}$ (2) (ii) $\frac{d}{dx} \frac{1}{(3x-2)^2} = \frac{d}{dx} (3x-2)^{-2}$ $= -2(3x-2)^{-3} \times 3$ $= \frac{-6}{(3x-2)^3}$ (2)		Students who used Gradient R. made many more errors.
(b) $\int (e^{2x} - \sin x) dx = \frac{1}{2} e^{2x} + \cos x + C$ (2) (c) $\int_4^5 \frac{1}{x-3} dx = [\log_e (x-3)]_4^5$ $= \log_e 2 - \log_e 1$ $= \log_e 2$ (2)		Check answer by Diff.... this avoids common error of sign or multiplier of 2.
(d) (i) $\frac{d}{dx} x^2 \ln x = \ln x \times (2x) + x^2 \times \frac{1}{x}$ $= 2x \ln x + x$ (2) (ii) $2x \ln x = \frac{d}{dx} x^2 \ln x - x$ $x \ln x = \frac{1}{2} \frac{d}{dx} x^2 \ln x - \frac{1}{2} x$ $\int x \ln x dx = \frac{1}{2} x^2 \ln x - \frac{1}{4} x^2 + C$ (2)		Use Product R. 1 mark for recognising $\int x \ln x dx = \frac{1}{2} x^2 \ln x \dots$

Mathematics: Question 3

Suggested Solutions

Marks
Awarded

Marker's Comments

(a) (i) $AC = \sqrt{8^2 + 3^2}$
 $= \sqrt{73}$ units.

(1)

1

Well done

(ii) $AC: \frac{y-2}{x+2} = \frac{5-2}{6+2} = \frac{3}{8}$

1

1 mk for gradient

$3x+6 = 8y-16$

1

1 mk correct method

$3x-8y+22=0$

(2)

(iii) $d = \left| \frac{3 \times 3 - 8 \times (-2) + 22}{\sqrt{3^2 + 8^2}} \right|$

1

Some did not
know formula.

$= \frac{47}{\sqrt{73}}$ units

1

(2)

(iv) Area $\triangle ABC = \frac{1}{2} \times \sqrt{73} \times \frac{47}{\sqrt{73}}$
 $= 23.5$ unit².

1

full mks if incorrect
pt (iii) used.

1

(2)

(b) (i) $x+y=5$ (1)

1

$2x-3y=-5$ (2)

(1) $\times 2: 2x+2y=10$ (3)

(2) $-(3): -5y=-15$

1

Some simple errors
in calculation.

$y=3$

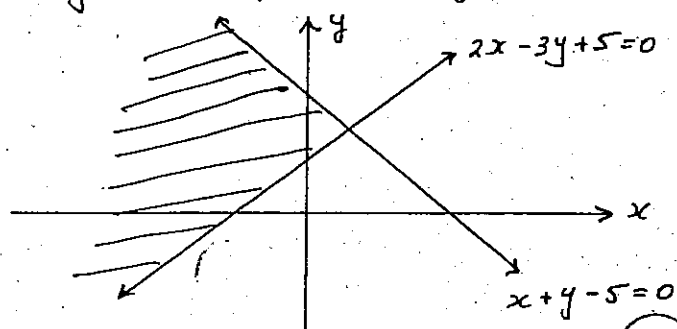
1

$\therefore x=2$ from (1)

Point of intersection is (2,3)

(3)

(ii) $x+y-5 \leq 0, 2x-3y+5 \leq 0$



1

1 mk for EACH
correct region.

1

Quite a few
careless errors
in shading

(2)

Mathematics: Question 4

Suggested Solutions

Marks
Awarded

Marker's Comments

(a) $2^{2x} - 7 \times 2^x - 8 = 0$

Let $u = 2^x$: $u^2 - 7u - 8 = 0$

$(u-8)(u+1) = 0$

$u = 8$ or $u = -1$

$2^x = 8$ or $2^x = -1$

$x = 3$ is the only solution.

(3)

(b) (i) $kx^2 + 4x + k$

$\Delta = 16 - 4k^2$

(1)

(ii) For no real roots, $\Delta < 0$

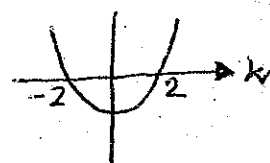
$16 - 4k^2 < 0$

$k^2 > 4$

$k < -2, k > 2$

(2)

Draw sketch to
solve $k^2 > 4$



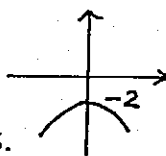
(c) $x^2 = -8(y+2)$

(i) Vertex $(0, -2)$

(ii) Focal length = 2 units.

(iii) Focus $(0, -4)$

(iv) Directrix: $y = 0$



(4x1)

Students who draw
sketch made for
fewer errors.

$x^2 = -4a(y-k)$

NB length $\Rightarrow +2$

OR.

$\left(\frac{-b + \sqrt{b^2 - 4ac}}{2a} - \frac{-b - \sqrt{b^2 - 4ac}}{2a} \right)$

$= \left(\frac{2\sqrt{b^2 - 4ac}}{2a} \right)^2$

(2)

(d) $ax^2 + bx + c = 0$

$(\alpha - \beta)^2 = \alpha^2 + \beta^2 - 2\alpha\beta$

$= \alpha^2 + 2\alpha\beta + \beta^2 - 4\alpha\beta$

$= (\alpha + \beta)^2 - 4\alpha\beta$

$= \left(-\frac{b}{a} \right)^2 - 4\left(\frac{c}{a} \right)$

$= \frac{b^2}{a^2} - \frac{4c}{a}$

$= \frac{b^2 - 4ac}{a^2}$

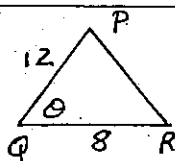
Mathematics: Question 5

Suggested Solutions

Marks
Awarded

Marker's Comments

(a) Using $\text{Area} = \frac{1}{2}ab \sin C$
 $\frac{1}{2} \times 12 \times 8 \times \sin \theta = 28.8$
 $\sin \theta = \frac{28.8}{48}$
 $= 0.6$
 $\theta = 37^\circ, 143^\circ$



✓

correct substitution
into area formula

(3)

✓✓

(✓) acute (✓) obtuse

(b) (i) $l = r\theta$
 $4 = 2.5\theta$
 $\theta = \frac{4}{2.5}$
 $= 1.6 \text{ radians.}$

(1)

✓

correct substitution

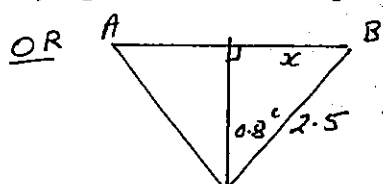
(ii) $AB^2 = 2.5^2 + 2.5^2 - 2 \times 2.5 \times 2.5 \cos 1.6^\circ$
 $= 12.5 - 12.5 \cos 1.6^\circ$ (COSINE RULE)
 $= 12.865$

$AB = 3.6 \text{ cm (1 d.p.)}$

(2)

✓

solution using
the cosine rule



$\frac{x}{2.5} = \sin 0.8^\circ$ ✓
 $x = 2.5 \sin 0.8$
 $= 1.793$ ✓

$\therefore AB = 3.6 \text{ cm (1 d.p.)}$

(2)

✓

solution making
a right angled
triangle.

(iii) $\text{Area} = \frac{1}{2}r^2(\theta - \sin \theta)$
 $= \frac{1}{2} \times 2.5^2 (1.6 - \sin 1.6^\circ)$
 $= 1.876 \text{ cm}^2 \text{ (3 d.p.)}$

(2)

✓

Note: calculator
must in radians mode
for this question

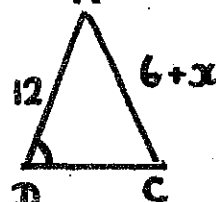
(c) (i) In $\triangle ABE$, $\triangle ACD$
 $\angle ABE = \angle ADC$ (given)
 $\angle BAE = \angle DAC$ (same angle)
 $\therefore \triangle ABE \parallel \triangle ACD$ (two angles equal)

(2)

✓

✓

equiangular



(ii) Let $BC = x \text{ cm}$

$\frac{6}{5} = \frac{12}{6+x}$ ✓

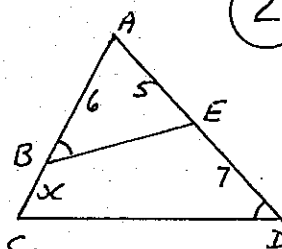
$36 + 6x = 60$

$6x = 24$

$x = 4$

$\therefore BC = 4 \text{ cm}$ ✓

(2)



if you draw the 2
similar triangles
next to each other
you will get the
ratio right.

Mathematics: Question 6.

Suggested Solutions	Marks Awarded	Marker's Comments
(a) $S_{16} = 288$, $T_6 = 8$ Let the series be $a + (a+d) + (a+2d) + \dots$ $a + 5d = 8$ (1) $\frac{16}{2} [2a + 15d] = 288$ i.e. $2a + 15d = 36$ (2) (1) $\times 2$: $2a + 10d = 16$ (3) (2) - (3): $5d = 20$ $d = 4$ From (1) $a = -12$ Series is $-12, -8, -4$	✓ ✓ ✓ ✓ (4) ✓	$T_n = a + (n-1)d$ $S_n = \frac{n}{2} [2a + (n-1)d]$
(b) $4 \times (1.2)^n > 560$ $(1.2)^n > 140$ $n \log 1.2 > \log 140$ $n > \frac{\log 140}{\log 1.2}$ $n > 27.103 \dots$ Least value of n is 28	✓ ✓ ✓ (3) ✓	"least positive integer > 27.103 is 28 NOT 27
(c) (i) First deposit is in for 10 years $A = 1500(1.04)^{20}$ $= \\$3286.68$ (ii) Total value $= 1500 \times (1.04)^{20} + 1500 \times (1.04)^{18} + \dots$ $\dots + 1500(1.04)^4 + 1500 \times (1.04)^2$ $= 1500(1.04)^2 \left[\frac{(1.04^2)^{10} - 1}{1.04^2 - 1} \right]$ $= 1500(1.04)^2 \left[\frac{1.04^{20} - 1}{1.04^2 - 1} \right]$ $= \\$23682.33$ using $S_n = \frac{a(r^n - 1)}{r - 1}$ where $r = (1.04)^2$	✓ (2) ✓ ✓ ✓ (3) ✓	$A = P \left(1 + \frac{r}{100}\right)^n$ where 8% p.a. compounded half yearly is 4% p. half year correct series $S_n = \frac{a[r^n - 1]}{r - 1}$ correct answer.

Mathematics: Question 7

Suggested Solutions	Marks Awarded	Marker's Comments								
(a) $y = 3x^2 - 8x + 2$ $\frac{dy}{dx} = 6x - 8$ At $(2, -2)$, $\frac{dy}{dx} = 6 \times 2 - 8 = 4$. Gradient of normal $= -\frac{1}{4}$. Normal: $y + 2 = -\frac{1}{4}(x - 2)$ $4y + 8 = -x + 2$ $x + 4y + 6 = 0$	1 1 1 (3)									
(b) $y = x^3 - 12x^2 + 48x + 50$ $\frac{dy}{dx} = 3x^2 - 24x + 48$ $= 3(x^2 - 8x + 16)$ $= 3(x - 4)^2$ At stationary point, $\frac{dy}{dx} = 0$ $\therefore 3(x - 4)^2 = 0 \therefore x = 4$ There is only one stationary point. $\frac{d^2y}{dx^2} = 6x - 24$ When $x = 4$, $\frac{d^2y}{dx^2} = 6 \times 4 - 24 = 0$ When $x = 3$, $\frac{d^2y}{dx^2} = 6 \times 3 - 24 = -6$ When $x = 5$, $\frac{d^2y}{dx^2} = 6 \times 5 - 24 = 6$ $\therefore x = 4$ is a point of inflexion since $\frac{d^2y}{dx^2} = 0$ and concavity changes.	1 1 1 1 1 1 (4)	It is <u>not</u> true that: $\frac{dy}{dx} = 0$ and $\frac{d^2y}{dx^2} = 0$ defines a H.P.I. (check with $y = x^4$) or first derivative test <table border="1"><tr><td>x</td><td>3</td><td>4</td><td>5</td></tr><tr><td>$\frac{d^2y}{dx^2}$</td><td>+3</td><td>0</td><td>+3</td></tr></table>	x	3	4	5	$\frac{d^2y}{dx^2}$	+3	0	+3
x	3	4	5							
$\frac{d^2y}{dx^2}$	+3	0	+3							
(c) (i) $4x + BH = 40$ $BH = 40 - 4x$ (ii) $A = x(40 - 4x)$ $= 40x - 4x^2$ $\frac{dA}{dx} = 40 - 8x$ $= 0$ when $x = 5$. $\frac{d^2A}{dx^2} = -8 < 0$ $\therefore$ Area is a maximum when $AB = 5\text{ m}$, $BH = 20\text{ m}$.	1 1 1 1 (4)	Must prove Max not just assume Read & carefully Area not required								

Mathematics: Question 8

Suggested Solutions

Marks
Awarded

Marker's Comments

(8) $y = 1 + \sin 2x$

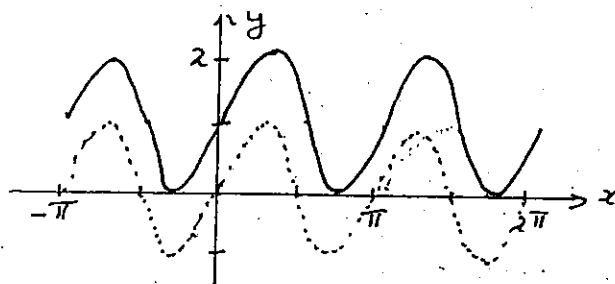
(i) Period = $\frac{2\pi}{2} = \pi$

(1)

1

Well done.

(ii)



(3)

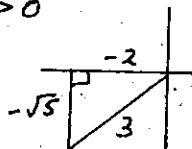
3

1 for shape
1 position
1 $-\pi \rightarrow 2\pi$
some quite messy.

(b) $\cos \theta = -\frac{2}{3}$, $\tan \theta > 0$

θ in 3rd quadrant.

$\sin \theta = -\frac{\sqrt{5}}{3}$



OR Use $\sin^2 \theta + \cos^2 \theta = 1$

$\sin^2 \theta + \frac{4}{9} = 1$

$\sin^2 \theta = \frac{5}{9} \therefore \sin \theta = \pm \frac{\sqrt{5}}{3}$

But $\sin \theta < 0 \therefore \sin \theta = -\frac{\sqrt{5}}{3}$

(2)

1

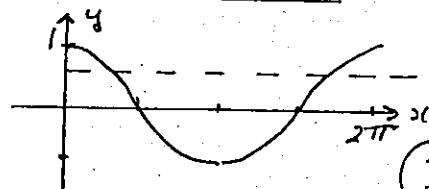
1 for correct $\sqrt{5}$

1 for correct 3

not well done

1

(c) (i) $y = \cos x$



(1)

1

axes NOT labeled
loss of mark.

(ii) $\cos x \leq \frac{\sqrt{3}}{2}$

If $\cos x = \frac{\sqrt{3}}{2}$, $x = \frac{\pi}{6}, \frac{11\pi}{6}$

For $\cos x \leq \frac{\sqrt{3}}{2}$, $\frac{\pi}{6} \leq x \leq \frac{11\pi}{6}$

(2)

1

1 for $x \geq \frac{\pi}{6}$

1 for $x \leq \frac{11\pi}{6}$

Extra INCORRECT
info loss of mk.

(d) (i) $\sin^2 \theta + \cos^2 \theta = 1$

Divide by $\cos^2 \theta$

(1)

1

Most OK.

(ii) $\int \tan^2 x \, dx = \int (\sec^2 x - 1) \, dx$
 $= \tan x - x + C$

(2)

1

Most have
 $\tan x$, $-x$ and
 $+C$ for both
marks.

1

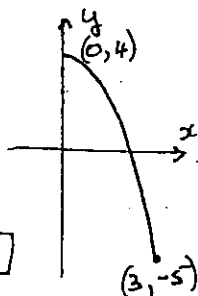
Mathematics: Question 9.

Suggested Solutions

Marks
Awarded

Marker's Comments

(a) $y = 4 - x^2$
 $V = \pi \int x^2 dy$
 $= \pi \int_{-5}^4 (4 - y) dy$
 $= \pi \left[4y - \frac{1}{2} y^2 \right]_{-5}^4$
 $= \pi \left[(16 - 8) - (-20 - 12\frac{1}{2}) \right]$
 Volume = $40.5 \pi \text{ unit}^3$.



(3)

$\frac{1}{1}$

- some rotated about the x axis.
 - generally OK.

(b) $y = 2x$, $y = x(x-3)$
 (i) $x^2 - 3x = 2x$
 $x^2 - 5x = 0$
 $x(x-5) = 0$
 $x = 0 \text{ or } x = 5$
 $\therefore x\text{-coord of A is } x = 5$

(1)

$\frac{1}{1}$

well done.

(ii) $A = \int_0^5 (y_1 - y_2) dx$
 $= \int_0^5 [2x - (x^2 - 3x)] dx$
 $= \int_0^5 (5x - x^2) dx$
 $= \left[\frac{5}{2} x^2 - \frac{1}{3} x^3 \right]_0^5$
 $= \left(\frac{125}{2} - \frac{125}{3} \right) - (0 - 0)$
 Area = $\frac{125}{6} \text{ unit}^2$

(3)

$\frac{1}{1}$

- many tried other ways with some success.
 - Some divided region into 3 areas. NOT SUCCESSFUL.

(c)(i) $\int_0^8 f(x) dx \doteq \frac{h}{3} [y_0 + 4y_1 + 2y_2 + 4y_3 + y_4]$
 $\doteq \frac{2}{3} [3 \cdot 2 + 4 \times 1 \cdot 4 + 2 \times 0 \cdot 6 + 4 \times 2 \cdot 2 + 3 \cdot 6]$
 $\doteq 14.93$

(2)

$\frac{1}{1}$

- most did this with little difficulty

(ii) $V = \pi \int y^2 dx$
 $\doteq \pi \times \frac{2}{3} [3 \cdot 2^2 + 4 \times 1 \cdot 4^2 + 2 \times 0 \cdot 6^2 + 4 \times 2 \cdot 2^2 + 3 \cdot 6^2]$
 $\doteq \frac{2\pi}{3} \times 51.2$
 Volume $\doteq 34.08 \pi \text{ or } 107 \text{ unit}^3$

(3)

$\frac{1}{1}$

- Many could not understand how to get a volume with this rule.
 - poorly done.

Mathematics: Question 10.

Suggested Solutions

Marks
Awarded

Marker's Comments

(a) $N = 2000 e^{0.005t}$
 (i) When $t=10$, $N = 2000 \times e^{0.05}$
 $= 2103$ (1)

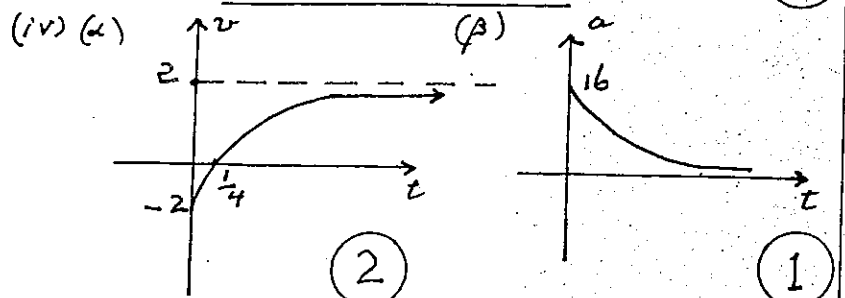
(ii) $\frac{dN}{dt} = 2000 \times e^{0.005t} \times 0.005$
 $= 10 e^{0.005t}$
 When $t=10$, $\frac{dN}{dt} = 10 \times e^{0.05}$
 $= 10.5$ (2)
 Colony is increasing at 10.5/minute.

(iii) When $N=4000$, $4000 = 2000 e^{0.005t}$
 $e^{0.005t} = 2$
 $0.005t = \ln 2$
 $t = \frac{\ln 2}{0.005}$
 $= 138.6$ (1)
 It takes 138.6 minutes to double.

(b) $x = 2t - \log_e(4t+1)$
 (i) $v = 2 - \frac{4}{4t+1}$
 $= \frac{8t+2-4}{4t+1}$
 $= \frac{8t-2}{4t+1}$
 $a = \frac{(4t+1) \times 8 - (8t-2) \times 4}{(4t+1)^2}$
 $= \frac{32t+8-32t+8}{(4t+1)^2}$
 $= \frac{16}{(4t+1)^2}$ (2)

(ii) When $t=0$, $v = -2 \text{ m/s}$.
 $a = 16 \text{ m/s}^2$ (2)

(iii) When $v=0$, $t = \frac{1}{4}$ (1)



Part (a) was well done.

b i) Need to show working for getting from $2 - \frac{4}{4t+1}$ to the required result $\frac{8t-2}{4t+1}$. Those that didn't, got no marks.

ii) well done

iii) when solving $\frac{8t-2}{4t+1} = 0$ many

made the error of having as the next line $8t-2 = 4t+1$ ($0 \times (4t+1) = 0$)

iv) no marks awarded if linear graphs were drawn for velocity and acceleration. correct shape and asymptote, $v=2$ needed to get full marks for velocity graph.